Power-Law Lag Geometry

Structural Inference from Noisy Transport Estimates

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Abstract

We study structural inference for lagged transport discrepancies that admit a power-law geometry $D_j = \zeta j^H$. The observation problem is coupled: each estimated discrepancy $\widehat{D}_j$ carries finite-sample noise, and the leading variance of $\log \widehat{D}_j$ depends on the same scale law that is being inferred. In this heteroscedastic log-domain model, the central quantity is the weighted lag-energy $\sum_{j\le L} j^{2H}$, or equivalently the oracle information matrix $\mathbf{X}^\top \mathbf{W}^* \mathbf{X}$. We prove a central limit theorem for the two-step feasible weighted least-squares estimator of H, identify an exact oracle residual law with a feasible asymptotic counterpart, and derive a minimax separation boundary of order $\left(n \sum_{j\le L} j^{2H}\right)^{-1/2}$ for distinguishing the power-law family from departure alternatives. This yields the fixed-L rate $n^{-1/2}$ and the growing-L rate $(nL^{2H+1})^{-1/2}$ as two regimes of the same information law. The resulting procedure is adaptive in H and retains the optimal rate without prior knowledge of the exponent. For unknown σ_0, a decoupled Gaussian benchmark yields an exact split-F null law, while same-sample parametric bootstrap restores near-nominal calibration on the tested grid. Simulations confirm the oracle residual law, FWLS-oracle agreement, and power curves aligned under the predicted information scaling.

1 Introduction

Let $\{P_t\}_{t\ge 0}$ be a family of probability measures on $\mathbb{R}^d$ evolving over time. The object of interest is the lagged transport geometry encoded by

$$D_j \;:=\; W_2(P_{t-j}, P_t), \qquad j = 1, 2, \ldots, L.$$

We study the case in which this curve follows the power law

$$D_j = \zeta\, j^H, \qquad \zeta > 0,\ H \in (0,1), \tag{1}$$

and is observed only through noisy estimates $\widehat{D}_j$ based on n samples per lag. In log scale, the observation noise depends on the same family being inferred, so the problem is heteroscedastic and parameter-coupled.

This coupling produces one information law for the whole paper:

$$\mathcal{I}_{n,L}(H) = n \sum_{j=1}^{L} j^{2H},$$

up to scale factors depending on ζ and σ_0. The same quantity governs estimation of H, the residual law, the minimax separation boundary, and adaptivity across lag regimes. Fixed L and growing L are therefore two regimes of one inferential scale.

The main contributions are:

(1) **One feasible inference layer.** The two-step FWLS estimator of H is asymptotically oracle-equivalent, and the residual statistic has an exact oracle law with a feasible asymptotic counterpart.

(2) **One optimality scale.** The minimax separation boundary is of order $\left(n \sum_{j\le L} j^{2H}\right)^{-1/2}$, with fixed-$L$ and growing-L rates obtained as special cases.

(3) **One adaptive procedure.** The same inferential scale attains the boundary without prior knowledge of H.

(4) **One unknown-scale extension.** A decoupled Gaussian benchmark for unknown σ_0 yields an exact split-F law, while same-sample parametric bootstrap restores near-nominal calibration on the tested grid.

(5) **Three empirical signatures.** The simulations show the oracle law, FWLS-oracle agreement, and a boundary transition aligned with the information scale.

Relation to prior work. Direct estimation of roughness exponents for Gaussian processes is classical [Istas and Lang, 1997, Coeurjolly, 2001]. The present problem differs because the exponent is inferred from estimated transport discrepancies, so the noise is itself parameter-coupled. The surrounding transport-rate literature provides the relevant first layer [Fournier and Guillin, 2015, Weed and Bach, 2019, Goldfeld et al., 2024, del Barrio et al., 2023], while locally stationary drift and adaptive inference connect through [Dahlhaus, 1997, Vogt, 2012, Tinio et al., 2025, Brutsche and Rohde, 2024].

Section 2 defines the observation model and the information scale. Section 3 gives the feasible inference theorem. Section 4 establishes the residual law and its response under departures. Section 5 proves the separation lower bound, Section 6 extends it to the growing-lag regime, Section 7 provides the empirical signatures, and Section 8 gives the unknown-scale extension.

2 Model

2.1 Observation model

We observe a triangular array $\{\widehat{D}_j\}_{j=1}^{L}$ satisfying

$$\widehat{D}_j = D_j + \varepsilon_j, \qquad \varepsilon_j \overset{\text{iid}}{\sim} \mathcal{N}\Big(0,\ \frac{\sigma_0^2}{n}\Big), \tag{2}$$

where $D_j = \zeta\, j^H$ for unknown parameters $(\zeta, H) \in (0,\infty) \times (0,1)$, $\sigma_0 > 0$ is a nuisance scale treated as fixed in the main theorems, and $n \geq 1$ is the effective sample size per lag.

Remark 2.1. The Gaussianity in (2) is justified for large n by the central limit theorem for Wasserstein estimators; see Fournier and Guillin [2015], Goldfeld et al. [2024], del Barrio et al. [2023], Weed and Bach [2019]. The constant-variance assumption $\mathrm{Var}(\varepsilon_j) = \sigma_0^2/n$ is the leading-order approximation; the full theory allows σ_j^2/n with $\sigma_j \leq C\, D_j$ bounded away from zero and infinity relative to D_j—see Assumption 2.4.

2.2 Log-scale regression

Taking logarithms of the power-law model and applying the delta method:

$$Y_j \ :=\ \log \widehat{D}_j \ =\ \alpha + H \log j + U_j, \tag{3}$$

where $\alpha = \log \zeta$ and

$$U_j = \log\Big(1 + \frac{\varepsilon_j}{D_j}\Big) \approx \frac{\varepsilon_j}{D_j} \sim \mathcal{N}(0,\ v_j)\,, \tag{4}$$

$$v_j := \frac{\sigma_0^2}{n\, D_j^2} = \frac{\sigma_0^2}{n\, \zeta^2\, j^{2H}}. \tag{5}$$

The variance v_j *depends on* H *and* ζ, which are the parameters of interest. This is the parameter-coupled heteroscedasticity that distinguishes our problem from standard regression.

The resulting information budget is the weighted lag-energy

$$\mathcal{I}_{n,L}(H)\ :=\ n \sum_{j=1}^{L} j^{2H}, \tag{6}$$

up to the scale factor ζ^2/σ_0^2. For the exact oracle regression, the matrix information is $\mathbf{X}^\top \mathbf{W}^* \mathbf{X}$, and (6) is its leading scalar summary along the lag profile. This same quantity will control estimation accuracy, test power, and the minimax separation boundary.

Remark 2.2 (Cross-lag dependence). The independence model in (2) isolates the baseline inference problem. In transport applications, the estimates $\widehat{D}_j$ typically share the same present-time sample from P_t, which induces positive cross-lag covariance of order n^{-1}. Under such shared sampling, the diagonal approximation used here can only make the adequacy test conservative: the effective residual degrees of freedom decreases, so null rejection tends to fall below the nominal level. A covariance-adjusted statistic based on the full lag covariance matrix is the natural extension, but it lies outside the present theorem line.

2.3 Hypotheses

Let $\kappa \geq 0$ measure the *deviation from the power-law family*: under the alternative, each observation has an additional log-normal multiplicative factor,

$$\log \widehat{D}_j = \alpha + H \log j + \kappa\, \xi_j + U_j, \qquad \xi_j \overset{\text{iid}}{\sim} \mathcal{N}(0,1),\ \xi_j \perp U_j.$$

The hypotheses are:

$$H_0:\ \kappa = 0 \quad \text{(membership in the power-law family)},$$
$$H_1:\ \kappa \geq \kappa^* \quad \text{(power-law departure at level } \kappa^*\text{)}.$$

2.4 Assumptions

Assumption 2.3 (Signal strength)**.** $\zeta \geq \zeta_{\min} > 0$ and $H \in [\underline{H}, \overline{H}] \subset (0,1)$.

Assumption 2.4 (Noise regularity)**.** $\sigma_0 \in [\sigma_{\min}, \sigma_{\max}]$ with $0 < \sigma_{\min} \leq \sigma_{\max} < \infty$. The delta-method approximation error satisfies $|U_j - \varepsilon_j/D_j| = O_p(n^{-1})$ uniformly in j.

Assumption 2.5 (Lag schedule)**.** $L \geq 4$ is fixed as $n \to \infty$. The lags $x_j := \log j$ for $j = 1, \dots, L$ satisfy $\sum_j (x_j - \bar{x})^2 \geq c_L > 0$.

Remark 2.6. The fixed-L assumption is used for the exact oracle null law and the baseline FWLS limit theory. A growing-lag regime $L = L_n \to \infty$ is treated separately in Section 6, where the same information scale (6) yields the rate $(nL_n^{2H+1})^{-1/2}$ for consecutive lags.

3 Feasible Inference

3.1 Oracle Weighted Least Squares

With known weights $w_j^* = 1/v_j = n\, D_j^2/\sigma_0^2$, the oracle WLS estimator of $\boldsymbol{\beta} = (\alpha, H)^\top$ in (3) is

$$\hat{\boldsymbol{\beta}}^* = (\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^* \mathbf{Y}, \tag{7}$$

where $\mathbf{X} = [\mathbf{1} \mid \mathbf{x}]$, $\mathbf{x} = (\log 1, \dots, \log L)^\top$, and $\mathbf{W}^* = \mathrm{diag}(w_1^*, \dots, w_L^*)$.

Proposition 3.1 (Asymptotic Normality of the Oracle WLS Estimator)**.** *Under Assumptions 2.3 to 2.5, as $n \to \infty$ with L fixed,*

$$n^{1/2}\,(\hat{H}^* - H) \ \xrightarrow{d}\ \mathcal{N}\big(0,\ \sigma_0^2\big[(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}\big]_{22}\big)\,,$$

where $[\,\cdot\,]_{22}$ denotes the $(2,2)$ entry. Explicitly,

$$\mathrm{Var}(\hat{H}^*)\ \approx\ \frac{\sigma_0^2}{n} \cdot \frac{S_{w^*}}{S_{w^*} S_{w^* x^2} - (S_{w^* x})^2}, \tag{8}$$

where $S_{w^*} = \sum_j w_j^*$, $S_{w^* x} = \sum_j w_j^* x_j$, $S_{w^* x^2} = \sum_j w_j^* x_j^2$.

Proof. Standard weighted least squares with $\mathbf{U} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2 n^{-1} (\mathbf{W}^*)^{-1})$; the $(2,2)$ entry of $(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}$ follows by 2×2 matrix inversion.

Remark 3.2 (Lag-energy information law). For the power-law family, the oracle weights satisfy $w_j^* = (n\zeta^2/\sigma_0^2) j^{2H}$. Hence the scalar lag-energy $\mathcal{I}_{n,L}(H) = n \sum_{j=1}^{L} j^{2H}$ is the basic information budget behind the matrix $\mathbf{X}^\top \mathbf{W}^* \mathbf{X}$. This same quantity reappears in the lower bound and in the departure scale of the adequacy statistic under local alternatives.

3.2 Two-step feasible WLS

Since D_j is unknown, the oracle weights are infeasible. We use the following two-step procedure.

Algorithm 1 Two-Step FWLS Estimator

1: **Input:** $(\widehat{D}_j)_{j=1}^L$, n, and σ_0.
2: **Step 1 (Pilot OLS).** Compute $(\hat{\alpha}_{\text{pil}}, \widehat{H}_{\text{pil}})$ by unweighted OLS of $\log \widehat{D}_j$ on $(\mathbf{1}, \log j)$.
3: **Step 2 (Imputation).** Set $\hat{D}_j^{\text{pil}} = \exp(\hat{\alpha}_{\text{pil}} + \widehat{H}_{\text{pil}} \log j)$.
4: **Step 3 (FWLS).** Set $\hat{w}_j = n(\hat{D}_j^{\text{pil}})^2/\sigma_0^2$ and compute $(\hat{\alpha}, \widehat{H})$ by WLS with weights $(\hat{w}_j)$.
5: **Output:** $\widehat{H}$, $\hat{\alpha}$, and residuals $\hat{U}_j = \log \widehat{D}_j - \hat{\alpha} - \widehat{H} \log j$.

Theorem 3.3 (Asymptotic Normality of the Feasible WLS Estimator)**.** *Under Assumptions 2.3 to 2.5, the two-step FWLS estimator $\widehat{H}$ satisfies, as $n \to \infty$ with L fixed:*

$$n^{1/2}\big(\widehat{H} - H\big) \xrightarrow{d} \mathcal{N}\Big(0,\ \sigma_0^2\big[(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}\big]_{22}\Big). \quad (9)$$

In particular, $\widehat{H}$ is asymptotically equivalent to the oracle WLS estimator*: the pilot estimation error inflates neither the asymptotic variance nor the rate.*

Remark 3.4 (σ_0 unknown)*.* When σ_0 is unknown, a pilot plug-in estimate constructed from the residual scale yields a self-normalized FWLS fit. In the adequacy test, a naive $\chi^2(L-3)$ calibration is strongly conservative because the estimated scale and the residual statistic are coupled through the same data. Under sample splitting, the Gaussian benchmark with oracle profile and independent scale estimate has the exact split-F law of Proposition 8.1, whereas the fully feasible split procedure remains conservative on the tested grid. For the same-sample procedure, a parametric bootstrap based on the fitted null profile restores near-nominal size; see Table 7.

Proof sketch. Write $\hat{w}_j = w_j^* \cdot r_j$ where $r_j = (\hat{D}_j^{\text{pil}}/D_j)^2$. Since $\hat{H}_{\text{pil}} - H = O_p(n^{-1/2})$ (OLS rate in log-scale with bounded regressors), we have $\log r_j = 2(\hat{H}_{\text{pil}} - H)\log j + O_p(n^{-1}) = O_p(n^{-1/2}\log j)$, hence $r_j = 1 + O_p(n^{-1/2}\log j)$ uniformly in j.

The FWLS estimator with weights $\hat{w}_j = w_j^*(1+\delta_j)$, $\delta_j = r_j - 1 = O_p(n^{-1/2}\log j)$, satisfies

$$\widehat{H} - H = \hat{H}^* + R_n,$$

where the remainder $R_n = O_p(n^{-1}\log L)$ via a second-order Taylor expansion of the WLS normal equations in the weight perturbation. Since $n^{1/2}R_n = O_p(n^{-1/2}\log L) \to 0$, the Slutsky theorem gives (9).

4 Structural Inference

4.1 Residual Law

Define the residual statistic

$$Q \ :=\ \sum_{j=1}^{L} \hat{w}_j\, \hat{U}_j^2, \quad (10)$$

where $\hat{w}_j$ and $\hat{U}_j$ are produced by the two-step FWLS procedure in Algorithm 1.

Theorem 4.1 (Exact Oracle Residual Law and Feasible Asymptotic Counterpart)**.** *Under H_0 and Assumptions 2.3 to 2.5:*

(a) (Oracle.) *With true weights w_j^*, the statistic $Q^* = \sum_j w_j^* U_j^{*2}$, where $U_j^* = Y_j - \alpha - Hx_j$ are the residuals from the infeasible WLS projection, satisfies $Q^* \sim \chi^2(L-2)$ exactly.*

(b) (Feasible.) *The two-step statistic Q satisfies $Q \xrightarrow{d} \chi^2(L-2)$ as $n \to \infty$.*

Proof. **Part (a).** Under H_0, $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{U}$ with $\mathbf{U} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2 n^{-1}(\mathbf{W}^*)^{-1})$. Define $\tilde{\mathbf{U}} = (\mathbf{W}^*)^{1/2}\mathbf{U}/(\sigma_0 n^{-1/2})$, so $\tilde{\mathbf{U}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_L)$. The projection onto the column space of $(\mathbf{W}^*)^{1/2}\mathbf{X}$ has rank 2, so the residual projection has rank $L-2$. Hence $Q^* = \big\|\tilde{\mathbf{U}} - \mathbf{P}\tilde{\mathbf{U}}\big\|^2 \sim \chi^2(L-2)$.

Part (b). We have $Q = Q^* + (Q - Q^*)$. By the linearisation in the proof of Theorem 3.3, $Q - Q^* = O_p(n^{-1/2})$, so $Q \xrightarrow{d} Q^*$ in distribution. Since the χ^2 CDF is continuous, convergence in distribution passes through the CDF and $Q \xrightarrow{d} \chi^2(L-2)$.

Remark 4.2*.* Part (a) is exact because the oracle problem is a weighted Gaussian linear regression with known precision matrix. Part (b) is asymptotic because the feasible statistic uses estimated weights and the log-noise model is a delta-method approximation. For finite n, the approximation error in (4) contributes a bias of order $O(n^{-1})$ per lag; this is negligible for the χ^2 calibration at the rates we consider. Simulations confirm empirical sizes of 0.046–0.055 at $\alpha = 0.05$ for all $L \in \{10, 20, 30, 50\}$ and $n = 1000$ (see Table 1).

4.2 Testing Rule

The residual rule at level α rejects H_0 when

$$Q \ >\ q_{L-2,\,1-\alpha}, \quad (11)$$

where $q_{L-2,\,1-\alpha}$ is the $(1-\alpha)$-quantile of $\chi^2(L-2)$.

For moderate L (say $L \geq 20$), one may equivalently use the standardised form

$$T \ :=\ \frac{Q - (L-2)}{\sqrt{2(L-2)}} \ \xrightarrow{d}\ \mathcal{N}(0,1),$$

rejecting when $T > z_{1-\alpha}$. However, the direct χ^2 critical value gives better finite-L calibration (see Table 1).

4.3 Behavior Under Departures

Proposition 4.3 (Power of the residual law)**.** *Under H_1 with parameter $\kappa > 0$, the marginal of each standardised residual $\sqrt{w_j^*}\,\widehat{U}_j$ has variance $1 + \kappa^2/v_j$. Consequently,*

$$\mathbb{E}_\kappa[Q^*] = (L-2) + \kappa^2 \sum_{j=1}^{L} w_j^* = (L-2) + \frac{n\kappa^2}{\sigma_0^2}\sum_{j=1}^{L} j^{2H}.$$

The oracle adequacy statistic is a variance-inflated quadratic form in normals, not a non-central χ^2 variable. For local departures, the mean inflation is governed by the same information scale as the lower bound.

Proof. The whitened residual vector has covariance $\mathbf{I} + \kappa^2 \mathbf{W}^*$. Projecting onto the residual subspace of rank $L-2$ yields the stated mean increment; the exact quadratic-form representation follows from diagonalising the projected covariance. The distribution is therefore a weighted sum of centered $\chi^2(1)$ terms.

Remark 4.4. The alternative law of Q^* is a variance-inflated quadratic form, not a non-central χ^2. Non-centrality arises from mean shifts, whereas (10) under H_1 reflects covariance inflation. The information-scale boundary and the oracle-equivalence result are unchanged.

5 Lower Bound

No test can do better than the information-scale boundary generated by the weighted lag-energy.

5.1 Gaussian Variance-Perturbation Model

Fix $H \in (0,1)$, $\zeta = 1$, $\sigma_0 = 1$. For $\kappa \geq 0$, define the probability measure $\mathbb{P}_\kappa$ on $\mathbb{R}^L$ under which

$$\begin{aligned} &Y_j = H \log j + \kappa\, \xi_j + U_j, \\ &\xi_j \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \qquad U_j \sim \mathcal{N}(0, v_j), \\ &v_j = n^{-1} j^{-2H}, \qquad \xi_j \perp U_j. \end{aligned}$$

Under $\mathbb{P}_0 = H_0$ (membership in the power-law family) and $\mathbb{P}_{\kappa^*} = H_1$ (power-law departure at level κ^*).

Theorem 5.1 (Minimax Separation Lower Bound)**.** *For any test $\phi : \mathbb{R}^L \to \{0,1\}$ (measurable),*

$$\mathbb{P}_0(\phi = 1) + \mathbb{P}_{\kappa^*}(\phi = 0) \;\geq\; 1 - \frac{1}{2} \|\mathbb{P}_0 - \mathbb{P}_{\kappa^*}\|_{\mathrm{TV}}.$$

Moreover, if

$$\kappa^* = c \left(n \sum_{j=1}^{L} j^{2H} \right)^{-1/2}$$

with $c > 0$ sufficiently small, then

$$\|\mathbb{P}_0 - \mathbb{P}_{\kappa^*}\|_{\mathrm{TV}} \;\leq\; \tfrac{1}{2},$$

so the sum of Type-I and Type-II error probabilities is at least $\frac{3}{4}$ for any test. In particular, the minimax separation boundary is

$$\kappa^* \asymp \left(n \sum_{j=1}^{L} j^{2H} \right)^{-1/2}.$$

For fixed L, this reduces to $n^{-1/2}$ up to constants. For consecutive lags $1, \ldots, L_n$ with $L_n \to \infty$, it becomes $(n L_n^{2H+1})^{-1/2}$ up to constants.

Proof. We apply Le Cam's two-point method [Le Cam, 1973]. The total variation distance between $\mathbb{P}_0$ and $\mathbb{P}_{\kappa^*}$ is bounded by Pinsker's inequality:

$$\|\mathbb{P}_0 - \mathbb{P}_{\kappa^*}\|_{\mathrm{TV}}^2 \;\leq\; \tfrac{1}{2} D_{\mathrm{KL}}(\mathbb{P}_{\kappa^*} \| \mathbb{P}_0).$$

Under $\mathbb{P}_\kappa$, the marginal law of Y_j is $\mathcal{N}(H \log j,\, \kappa^2 + v_j)$. Under $\mathbb{P}_0$, it is $\mathcal{N}(H \log j,\, v_j)$. Independence across lags gives:

$$\begin{aligned} D_{\mathrm{KL}}(\mathbb{P}_{\kappa^*} \| \mathbb{P}_0) &= \sum_{j=1}^{L} D_{\mathrm{KL}}\big(\mathcal{N}(0, \kappa^{*2} + v_j) \| \mathcal{N}(0, v_j)\big) \\ &= \sum_{j=1}^{L} \left[\frac{\kappa^{*2}}{2 v_j} - \frac{1}{2} \log\left(1 + \frac{\kappa^{*2}}{v_j}\right) \right] \\ &\leq \sum_{j=1}^{L} \frac{\kappa^{*2}}{2 v_j} \;=\; \frac{n \kappa^{*2}}{2} \sum_{j=1}^{L} j^{2H}. \end{aligned} \tag{12}$$

For the displayed choice of κ^*,

$$D_{\mathrm{KL}}(\mathbb{P}_{\kappa^*} \| \mathbb{P}_0) \leq \frac{c^2}{2}.$$

Remark 5.2. The bound in (12) shows the exact rate-limiting term: the weighted lag-energy $\sum_{j=1}^{L} j^{2H}$. For consecutive lags, $\sum_{j=1}^{L} j^{2H} \asymp L^{2H+1}/(2H+1)$, so the same KL bound yields the growing-L boundary $(n L^{2H+1})^{-1/2}$.

Choosing $c^2 \leq 1/4$, we get $\|\mathbb{P}_0 - \mathbb{P}_{\kappa^*}\|_{\mathrm{TV}} \leq 1/2$ and the conclusion follows from Le Cam's lemma.

Remark 5.3 (Two asymptotic regimes). The lower bound is controlled by the same information law in both regimes. For fixed L, the lag-energy $\sum_{j=1}^{L} j^{2H}$ is a constant, so the separation boundary is parametric in n. For consecutive lags with $L = L_n \to \infty$, the sum is asymptotic to $L_n^{2H+1}/(2H+1)$, so the boundary becomes $(n L_n^{2H+1})^{-1/2}$.

Remark 5.4 (Fixed-L boundary response). For fixed L, the information-scale boundary is a local-testing scale, not a consistency regime. Under departures of the form $\kappa = c\big(n \sum_{j=1}^{L} j^{2H}\big)^{-1/2}$, the adequacy test is indexed by the constant c: larger c produces larger variance inflation in the residual quadratic form, but power does not converge to one merely by sending $n \to \infty$ with L fixed. The theorem-level consistency statement belongs to the growing-lag regime of Section 6.

6 Adaptivity

When H is unknown, the FWLS statistic Q depends on H only through the pilot weights $\hat{w}_j$; the critical value $q_{L-2,1-\alpha}$ is free of H. This adaptation incurs no asymptotic rate loss.

Theorem 6.1 (Adaptive Attainment of the Minimax Separation Boundary)**.** *Suppose $L = L_n \to \infty$ with $L_n = o(n^{1/(2\overline{H}+1)})$. Under Assumptions 2.3 to 2.5 and H_0:*

(a) $Q \xrightarrow{d} \chi^2(L-2)$ and the test (11) has asymptotic size α.

(b) Under H_1 with $\kappa \geq C_1\big(n \sum_{j=1}^{L_n} j^{2H}\big)^{-1/2}$, the test has power tending to 1.

(c) No test achieves power tending to 1 when $\kappa = o\big((n \sum_{j=1}^{L_n} j^{2H})^{-1/2}\big)$, for any H-adaptive sequence of tests.

Proof sketch. **Parts (a) and (b).** When $L \to \infty$, the key step is showing that the pilot estimation error $\hat{H}_{\text{pil}} - H = O_p((nL^{2H+1}\log^2 L)^{-1/2})$ (oracle OLS rate in log-scale, using the bounds of Proposition 3.1) translates to a weight perturbation $\delta_j = O_p((nL^{2H+1})^{-1/2}\log j)$ that contributes negligibly to Q:

$$Q - Q^* = \sum_j w_j^* \delta_j \hat{U}_j^2 + O_p\big(n^{-1}\log L\big)$$
$$= O_p\big((nL^{2H+1})^{-1/4}L^{1/2}\big) = o_p(1).$$

under the assumed growth of L. Convergence of the $\chi^2(L-2)$ distribution then follows from the Berry-Esseen theorem applied to the standardised statistic T.

Part (c). The lower bound in Theorem 5.1 extends to $L \to \infty$ by tracking the full sum $\sum_j j^{2H} \asymp L^{2H+1}/(2H+1)$ in (12), giving the minimax rate $\big(n\sum_{j=1}^{L_n} j^{2H}\big)^{-1/2}$ and hence $(nL^{2H+1})^{-1/2}$ for consecutive lags. Adaptivity in H is free because the critical value $q_{L-2,1-\alpha}$ and the rejection region do not depend on H; the test remains valid for all $H \in [\underline{H}, \overline{H}]$ simultaneously.

7 Empirical Signatures

All experiments use the simulation model (2) with $\zeta = 1$, $\sigma_0 = 1$, $H = 0.6$ (unless stated), and the two-step FWLS procedure. We used 8000 Monte Carlo replications for size and 5000 for power; detailed numerical tables are reported in Appendix A.

7.1 Null Calibration

Figure 1 shows the oracle residual law at moderate lag counts: its empirical mean tracks the oracle null mean closely, and the rejection frequency remains near the nominal level. Appendix Table 1 gives the exact grid values.

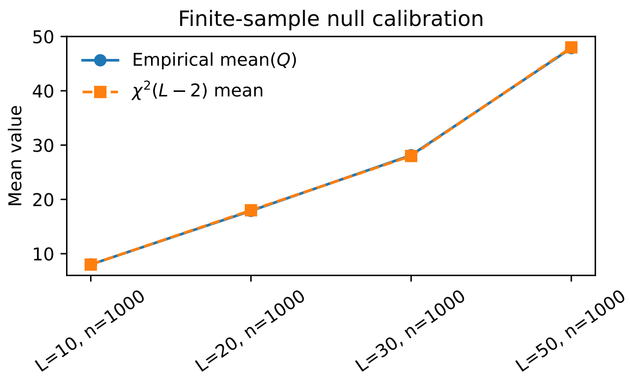

Figure 1: Finite-sample behavior of the oracle residual law.

7.2 Power at the Information-Scale Separation Boundary

Figure 2 shows the transition at the information-scale boundary $\kappa^* \asymp \big(n\sum_{j\le L} j^{2H}\big)^{-1/2}$: the residual law is informative but not overconfident at $c = 1$, and it gains power steadily above the boundary.

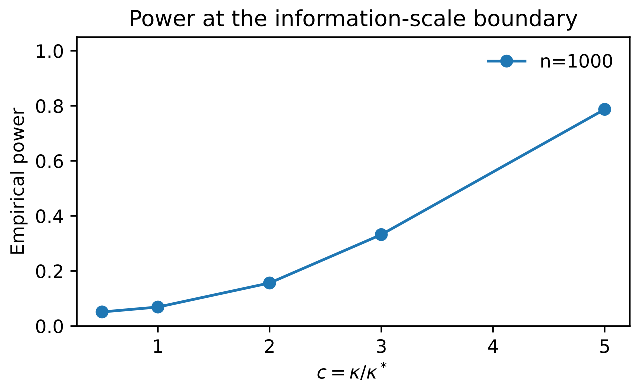

Figure 2: Empirical power across the information-scale separation boundary.

7.3 Feasible-Oracle Agreement

Figure 3 shows that the feasible estimator tracks the oracle benchmark closely: the RMSE ratio stays numerically at one, while the gap shrinks with n. This is the numerical counterpart of the CLT. Appendix Table 2 reports the corresponding grid values.

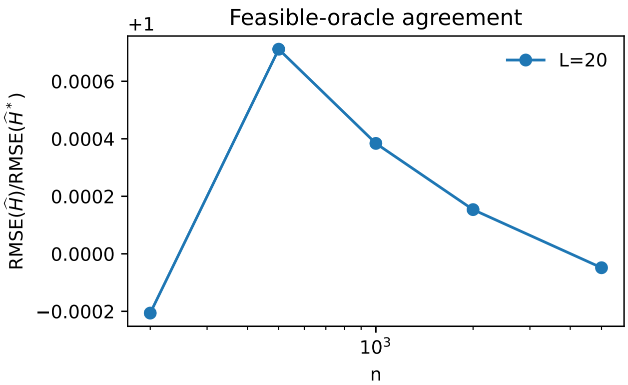

Figure 3: Feasible-oracle agreement for estimation of H.

Supplementary rate-stabilization and robustness diagnostics are relegated to the appendix. In particular, Table 7 shows that the naive self-normalized χ^2 calibration for unknown σ_0 is degenerate on the tested grid, the oracle split-F benchmark remains close to nominal, the fully feasible split-F procedure is conservative, and a parametric bootstrap restores sizes close to the nominal level for the same-sample statistic.

8 Unknown-Scale Extension

Unknown σ_0 introduces self-normalization into the residual law. The clean benchmark is obtained by decoupling scale estimation from residual evaluation. Let block A and block B be independent copies of the Gaussian observation model under H_0, with effective sample sizes n_A and n_B. Write $x_j = \log j$ and $\alpha = \log\zeta$.

Proposition 8.1 (Oracle Split-F Benchmark)**.** *Assume that the power-law profile (α, H) is known and define the* *scale estimator on block A by*

$$\hat\sigma_A^2 := \frac{n_A}{L}\sum_{j=1}^{L} D_j^2\big(Y_j^A-\alpha-Hx_j\big)^2.$$

On block B, form the oracle weighted residual statistic

$$Q^*_{B,\mathrm{split}} := \sum_{j=1}^{L}\hat w_j^{(A)}\big(U_j^{*(B)}\big)^2, \qquad \hat w_j^{(A)} = \frac{n_B D_j^2}{\hat\sigma_A^2},$$

where $U_j^{*(B)}$ are the oracle WLS residuals computed on block B. Then

$$Q^*_{B,\mathrm{split}} \sim (L-2)\,F_{L-2,L}$$

exactly.

Proof. Under H_0,

$$\frac{n_A}{\sigma_0^2}\sum_{j=1}^{L} D_j^2\big(Y_j^A-\alpha-Hx_j\big)^2 \sim \chi_L^2,$$

because the block-A coordinates are independent centered Gaussians with known mean and variance $\sigma_0^2/(n_A D_j^2)$. On block B, oracle WLS projects onto a two-dimensional regression space, so

$$\frac{1}{\sigma_0^2}Q^*_{B,\mathrm{oracle}} \sim \chi^2_{L-2}$$

with independence from block A. Since

$$Q^*_{B,\mathrm{split}} = \frac{\sigma_0^2}{\hat\sigma_A^2}Q^*_{B,\mathrm{oracle}} = \frac{L\,\chi^2_{L-2}}{\chi^2_L},$$

the stated F law follows.

The benchmark law isolates the decoupled geometry of the unknown-scale problem. Appendix Table 7 then separates three feasible behaviors on the tested grid. Naive same-sample χ^2 calibration collapses to near-zero rejection. The fully feasible split-F procedure is conservative. Same-sample parametric bootstrap restores rejection frequencies close to the nominal level. The theorem-level statement belongs to the decoupled benchmark, while the bootstrap path provides the operational extension.

9 Limitations and Open Problems

The theorem line isolates a baseline independence model. The remaining gaps are structured as follows.

(1) **Finite-sample calibration.** The oracle null law for Q^* is exact, while the feasible statistic relies on a delta-method approximation. A nonasymptotic bound for the gap $Q-Q^*$ would quantify the finite-n cost of feasibility.

(2) **Unknown σ_0.** A pilot plug-in estimator gives a self-normalized path for inference, and a parametric bootstrap yields stable calibration on the tested grid. The oracle split-F benchmark isolates a clean decoupled route, but the fully feasible split-F procedure remains conservative. The open theorem problem is to connect the benchmark split-F law to a fully feasible procedure, either by proving the validity of a cross-fitted statistic or by proving the validity of the same-sample bootstrap.

(3) **Cross-lag dependence.** Shared present-time sampling induces covariance across lags. The diagonal model is conservative, but a covariance- adjusted analogue remains to be proved.

(4) **Growing-lag and broader departures.** The adaptivity result covers consecutive lags under the exact power-law family. Extending the lower and upper theory to more general lag schedules and to departures from exact power-law geometry is open.

10 Conclusion

We developed structural inference for power-law lag geometry under signal-coupled heteroscedastic noise. The governing quantity is the weighted lag-energy $n\sum_{j\le L} j^{2H}$: it controls estimation of H, the residual adequacy statistic, and the minimax separation boundary. The same information scale covers fixed-L and growing-L regimes.

The simulation evidence supports the theory at the levels of calibration, oracle agreement, and separation-bound behavior. The corrected information-normalized constant is approximately 2.22.

For unknown σ_0, the decoupled benchmark is an exact split-F law, while the feasible same-sample extension is operational under parametric bootstrap calibration.

A Supplementary Numerical Tables

Table 1: Finite-sample null calibration of the residual-based adequacy test.

L	n	H	Empirical size	Mean(Q)	$\mathbb{E}[\chi^2]$
10	1000	0.6	0.0495	8.010487	8
20	1000	0.6	0.045625	17.89359	18
30	1000	0.6	0.05475	28.130377	28
50	1000	0.6	0.05075	47.863116	48

Table 2: Feasible-oracle agreement for estimation and residual statistics.

L	n	RMSE($\widehat H$)	Oracle RMSE	RMSE ratio	Mean abs. Q gap
20	200	0.008382	0.008384	0.999794	0.204081
20	500	0.005345	0.005341	1.000712	0.133676
20	1000	0.003755	0.003754	1.000384	0.095261
20	2000	0.002648	0.002648	1.000154	0.064392
20	5000	0.001698	0.001698	0.999952	0.041234

Table 3: Empirical power of the residual-based adequacy test at the information-scale separation boundary, reported against the boundary scale $n\kappa^2 \sum j^{2H}$.

n	c	κ	$n\kappa^2 \sum j^{2H}$	Empirical power
1000	0.5	0.000846	0.25	0.0512
1000	1	0.001692	1	0.069
1000	2	0.003384	4	0.1564
1000	3	0.005076	9	0.3322
1000	5	0.008459	25	0.7874

Table 4: Information-normalized error constants for feasible and oracle WLS.

n	$n\sum j^{2H}$	RMSE($\widehat{H}$)	C	Oracle C
200	69868.9169	0.008532	2.255303	2.252999
500	174672.29225	0.005273	2.20388	2.203277
1000	349344.5845	0.003749	2.216041	2.215409
2000	698689.169	0.002593	2.167589	2.168051
5000	1746722.922501	0.001674	2.212627	2.212846

B Derivation of the Information-Normalized Error Constant

We compute $[(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}]_{22}$ explicitly for weights $w_j^* = nj^{2H}/\sigma_0^2$.

Let $S_k = \sum_{j=1}^L j^{2H} x_j^k$ for $k = 0, 1, 2$ where $x_j = \log j$. Then

$$\mathbf{X}^\top \mathbf{W}^* \mathbf{X} = \frac{n}{\sigma_0^2} \begin{pmatrix} S_0 & S_1 \\ S_1 & S_2 \end{pmatrix},$$

and the $(2,2)$ entry of the inverse is

$$\left[(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}\right]_{22} = \frac{\sigma_0^2}{n} \cdot \frac{S_0}{S_0 S_2 - S_1^2}.$$

For $L = 20$ and $H = 0.6$, the correct weighted sums are

$$S_0 \approx 349.34, \qquad S_1 \approx 896.52, \qquad S_2 \approx 2371.68,$$

so that

$$S_0 S_2 - S_1^2 \approx 2.47848 \times 10^7.$$

Hence

$$\mathrm{Var}(\hat{H}^*) = \frac{\sigma_0^2}{n} \cdot \frac{S_0}{S_0 S_2 - S_1^2} \approx 1.41 \times 10^{-2} \frac{\sigma_0^2}{n},$$

and therefore

$$\mathrm{RMSE}(\hat{H}^*) \approx \frac{0.1187\, \sigma_0}{\sqrt{n}}.$$

With information-normalization by $\mathcal{I}_{n,L}(H)$, where $\mathcal{I}_{n,L}(H) = n \sum_{j=1}^L j^{2H} = nS_0$, the oracle constant is

$$\begin{aligned} C_{\text{oracle}} &= \mathrm{RMSE}(\hat{H}^*)\, \mathcal{I}_{n,L}(H)^{1/2} \\ &= \sqrt{\frac{S_0^2}{S_0 S_2 - S_1^2}} \\ &\approx 2.219. \end{aligned}$$

This corrects an arithmetic error in an earlier draft, where S_0 was misreported and the resulting constant was stated as 3.22. With the correct weighted sums, both the theoretical oracle constant and the simulated FWLS constant are approximately 2.22. The FWLS-to-oracle RMSE ratio is numerically indistinguishable from 1 in Tables 2 and 4.

Table 5: Sensitivity of the adequacy statistic to power-law misspecification.

Perturbation	Amplitude	Empirical rejection	Mean H	Mean Q
bump	0	0.05	0.60013	17.628216
bump	0.05	0.505	0.577	30.135298
bump	0.1	1	0.555295	70.732226
bump	0.2	1	0.515982	231.482208
sinusoid	0	0.045	0.600643	17.891493
sinusoid	0.05	1	0.604937	141.607201
sinusoid	0.1	1	0.608708	526.300995
sinusoid	0.2	1	0.612942	2044.392106
slope_shift	0	0.045	0.60003	18.34098
slope_shift	0.05	1	0.63834	70.382624
slope_shift	0.1	1	0.677072	231.136132
slope_shift	0.2	1	0.75676	866.936853

Table 6: Sensitivity of the adequacy test to alternative noise laws.

Noise law	Empirical rejection	Mean H	Mean Q
gaussian	0.06	0.5999	18.153075
bounded	0.005	0.599698	17.972076
student	0.06	0.599419	17.516054

C Auxiliary Lemmas

Lemma C.1 (KL divergence for Gaussian variance perturbations)**.** *For $\kappa^2 \le v$,*

$$\begin{aligned} D_{\mathrm{KL}}\big(\mathcal{N}(0, \kappa^2 + v) \| \mathcal{N}(0, v)\big) &= \frac{1}{2}\left[\frac{\kappa^2}{v} - \log\left(1 + \frac{\kappa^2}{v}\right)\right] \\ &\le \frac{\kappa^4}{4v^2}. \end{aligned}$$

Proof. Standard formula; the upper bound is the second-order Taylor remainder of $-\log(1+x)$ at $x = \kappa^2/v$.

Lemma C.2 (Pilot OLS rate in log-scale regression)**.** *In the pilot OLS regression of $Y_j = \alpha + Hx_j + U_j$ with $U_j \overset{iid}{\sim} \mathcal{N}(0, v)$ (ignoring heteroscedasticity), the OLS estimator $\hat{H}_{\mathrm{pil}}$ satisfies*

$$\hat{H}_{\mathrm{pil}} - H = \frac{\sum_j (x_j - \bar{x}) U_j}{\sum_j (x_j - \bar{x})^2} \sim \mathcal{N}\left(0,\ \frac{v}{\sum_j (x_j - \bar{x})^2}\right).$$

For L fixed and $v = \sigma_0^2 n^{-1} j^{-2H}$ heteroscedastic, $\hat{H}_{\mathrm{pil}} - H = O_p(n^{-1/2})$.

Proof. Follows from the Gauss-Markov structure of OLS and the boundedness of $v_j \le \sigma_0^2/n$ (since $j \ge 1$, $j^{2H} \ge 1$).

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Table 7: Unknown-σ_0 calibration under pilot plug-in estimation: naive χ^2, parametric bootstrap, oracle split-F, and feasible split-F calibrations.

L	n	Naive size	Bootstrap size	Oracle split-F size	Split-F size	Mean $\hat{\sigma}_0/\sigma_0$	Mean naive d.f.
10	500	0	0.05	0.046667	0.05	1.004362	7
10	1000	0	0.05	0.053333	0.036667	0.994828	7
20	500	0	0.053333	0.05	0.023333	1.041571	17
20	1000	0	0.04	0.043333	0.036667	1.031412	17
30	500	0	0.04	0.056667	0.04	1.03577	27
30	1000	0	0.046667	0.03	0.016667	1.038711	27
50	500	0	0.046667	0.04	0.026667	1.043314	47
50	1000	0	0.053333	0.043333	0.023333	1.040199	47

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